

Linear Algebra 1

Problem Set — Bases, Subspaces, and Vector-Space Isomorphisms

Problem 1 — Which modified lists are bases?

Exact statement. Suppose that the vectors $e_1, \dots, e_n$ form a basis for a real vector space V . Which of the following are also bases?

- (a) $e_1 + e_2, e_2 + e_3, \dots, e_{n-1} + e_n, e_n$;
- (b) $e_1 + e_2, e_2 + e_3, \dots, e_{n-1} + e_n, e_n + e_1$;
- (c) $e_1 - e_n, e_2 + e_{n-1}, \dots, e_n + (-1)^n e_1$.

Solution. Write the proposed lists as $v_1, \dots, v_n$.

(a) Let $v_k = e_k + e_{k+1}$ for $1 \leq k \leq n-1$ and $v_n = e_n$. Then we can recover the original basis:

$$e_n = v_n, \quad e_{n-1} = v_{n-1} - e_n, \quad e_{n-2} = v_{n-2} - e_{n-1}, \quad \dots, \quad e_1 = v_1 - e_2.$$

Hence $v_1, \dots, v_n$ span V and are linearly independent; they form a basis.

(b) Let $v_k = e_k + e_{k+1}$ for $1 \leq k \leq n-1$ and $v_n = e_n + e_1$. Consider the alternating sum

$$S := v_1 - v_2 + v_3 - \dots + (-1)^{n-1} v_n.$$

A direct cancellation shows:

$$S = \begin{cases} 0, & n \text{ even,} \\ 2e_1, & n \text{ odd.} \end{cases}$$

If n is even, this gives a nontrivial linear dependence, so the list is not a basis. If n is odd, then $e_1 = \frac{1}{2}S$ lies in $\text{span}(v_1, \dots, v_n)$, and then

$$e_2 = v_1 - e_1, \quad e_3 = v_2 - e_2, \quad \dots, \quad e_n = v_{n-1} - e_{n-1},$$

so all e_i lie in the span; hence $v_1, \dots, v_n$ span V . Since there are n vectors spanning an n -dimensional space, they form a basis. Therefore (b) is a basis iff n is odd.

(c) Interpret the pattern as

$$v_k := e_k + (-1)^k e_{n+1-k} \quad (1 \leq k \leq n).$$

Pair indices k and $m := n+1-k$. Then

$$v_k = e_k + (-1)^k e_m, \quad v_m = e_m + (-1)^m e_k.$$

If n is odd, then $(-1)^m = (-1)^{n+1-k} = (-1)^{n+1}(-1)^k = (+1)(-1)^k = (-1)^k$, so

$$v_k - (-1)^k v_m = 0$$

for each pair, giving a nontrivial dependence; hence not a basis.

If n is even, then $(-1)^m = (-1)^{n+1-k} = (-1)^{n+1}(-1)^k = -(-1)^k$, so

$$v_k = e_k + s e_m, \quad v_m = e_m - s e_k, \quad (s := (-1)^k).$$

Solving yields

$$e_k = \frac{v_k - s v_m}{2}, \quad e_m = \frac{s v_k + v_m}{2}.$$

Thus every e_i lies in $\text{span}(v_1, \dots, v_n)$, so the v_i span V and hence form a basis. Therefore (c) is a basis iff n is even.

Problem 2 — Unions and distributive-looking identities

Exact statement. Let T, U, W be subspaces of V .

(i) Show that $T \cup U$ is a subspace of V only if either $T \subseteq U$ or $U \subseteq T$.

(ii) Give explicit counterexamples to:

(a) $T + (U \cap W) = (T + U) \cap (T + W)$;

(b) $(T + U) \cap W = (T \cap W) + (U \cap W)$.

(iii) Show that each equality in (ii) can be replaced by a valid inclusion of one side in the other.

Solution. (i) If $T \subseteq U$ or $U \subseteq T$, then $T \cup U$ equals the larger subspace, hence is a subspace. Conversely, assume $T \cup U$ is a subspace and neither contains the other. Choose $t \in T \setminus U$ and $u \in U \setminus T$. Since $T \cup U$ is a subspace, it is closed under addition, so $t + u \in T \cup U$. If $t + u \in T$, then $u = (t + u) - t \in T$, contradiction. If $t + u \in U$, then $t = (t + u) - u \in U$, contradiction. Hence one must contain the other.

(ii)(a) In $V = \mathbb{R}^2$, let

$$T = \text{span}(e_1), \quad U = \text{span}(e_2), \quad W = \text{span}(e_1 + e_2).$$

Then $U \cap W = \{0\}$, so $T + (U \cap W) = T$. But $T + U = \mathbb{R}^2$ and $T + W = \mathbb{R}^2$, hence

$$(T + U) \cap (T + W) = \mathbb{R}^2 \neq T.$$

(ii)(b) In $V = \mathbb{R}^2$, let

$$T = \text{span}(e_1 + e_2), \quad U = \text{span}(e_1 - e_2), \quad W = \text{span}(e_1).$$

Then $T + U = \mathbb{R}^2$, so $(T + U) \cap W = W$. But $T \cap W = \{0\}$ and $U \cap W = \{0\}$, so

$$(T \cap W) + (U \cap W) = \{0\} \neq W.$$

(iii) For (a), if $x \in T + (U \cap W)$ then $x = t + z$ with $t \in T$ and $z \in U \cap W$. Thus $x \in T + U$ and $x \in T + W$, so

$$T + (U \cap W) \subseteq (T + U) \cap (T + W).$$

For (b), if $x \in (T \cap W) + (U \cap W)$ then $x = t + u$ with $t \in T \cap W$ and $u \in U \cap W$. Then $t, u \in W$ so $x \in W$, and also $t \in T$, $u \in U$ so $x \in T + U$. Hence

$$(T \cap W) + (U \cap W) \subseteq (T + U) \cap W.$$

Problem 3 — Decide if $V \cong W$ and exhibit an isomorphism if so

Exact statement. For each pair (V, W) of vector spaces over $\mathbb{R}$, either give an isomorphism $V \rightarrow W$ or show that no such isomorphism can exist. Here $P(\mathbb{R})$ denotes polynomial functions $\mathbb{R} \rightarrow \mathbb{R}$ and $C[a, b]$ denotes continuous real-valued functions on $[a, b]$.

- (a) $V = \mathbb{R}^4$, $W = \{x \in \mathbb{R}^5 : x_1 + x_2 + x_3 + x_4 + x_5 = 0\}$.
- (b) $V = \mathbb{R}^5$, $W = \{p \in P : \deg p \leq 5\}$.
- (c) $V = C[0, 1]$, $W = C[-1, 1]$.
- (d) $V = C[0, 1]$, $W = \{f \in C[0, 1] : f(0) = 0, f \text{ continuously differentiable}\}$.
- (e) $V = \mathbb{R}^2$, $W = \{\text{real solutions of } \ddot{x}(t) + x(t) = 0\}$.
- (f) $V = \mathbb{R}^4$, $W = C[0, 1]$.
- (g) (Harder) $V = P(\mathbb{R})$, $W = \mathbb{R}^{\mathbb{N}}$.

Solution.

(a) Define

$$\phi : \mathbb{R}^4 \rightarrow W, \quad \phi(a, b, c, d) = (a, b, c, d, -(a + b + c + d)).$$

Then ϕ is linear, injective, and surjective (projection onto the first four coordinates is an inverse). Hence $V \cong W$.

(b) W has basis $\{1, x, x^2, x^3, x^4, x^5\}$, so $\dim W = 6$, while $\dim \mathbb{R}^5 = 5$. Thus no isomorphism exists.

(c) Let $\alpha : [-1, 1] \rightarrow [0, 1]$ be $\alpha(t) = \frac{t+1}{2}$. Define

$$F : C[0, 1] \rightarrow C[-1, 1], \quad (Ff)(t) = f(\alpha(t)).$$

Then F is linear and bijective with inverse

$$(F^{-1}g)(x) = g(2x - 1).$$

Hence $C[0, 1] \cong C[-1, 1]$.

(d) Define

$$T : C[0, 1] \rightarrow W, \quad (Tf)(x) = \int_0^x f(t) dt.$$

Then $Tf \in C^1[0, 1]$, $(Tf)(0) = 0$, and $(Tf)' = f$. Thus T is linear. If $Tf = 0$, then differentiating gives $f = 0$, so T is injective. If $g \in W$, then $g' = h \in C[0, 1]$ and

$$g(x) = g(0) + \int_0^x g'(t) dt = \int_0^x h(t) dt = Th(x),$$

so T is surjective. Hence $V \cong W$ with inverse $g \mapsto g'$.

(e) The solution set of $\ddot{x} + x = 0$ is

$$W = \{x(t) = a \cos t + b \sin t : a, b \in \mathbb{R}\}.$$

Define

$$\psi : \mathbb{R}^2 \rightarrow W, \quad \psi(a, b)(t) = a \cos t + b \sin t.$$

This map is linear and bijective, so $V \cong W$.

(f) $\mathbb{R}^4$ is finite-dimensional. But $C[0, 1]$ is infinite-dimensional since $\{1, x, x^2, \dots\}$ is linearly independent in $C[0, 1]$. Therefore no isomorphism exists.

(g) $P(\mathbb{R})$ has countable basis $\{1, x, x^2, \dots\}$, hence $\dim P(\mathbb{R}) = \aleph_0$. In $\mathbb{R}^{\mathbb{N}}$, for each $r \in \mathbb{R}$ define

$$v_r := (1, r, r^2, r^3, \dots).$$

If $r_1, \dots, r_k$ are distinct and $\sum_{i=1}^k a_i v_{r_i} = 0$, then looking at coordinates $n = 0, 1, \dots, k-1$ gives

$$\sum_{i=1}^k a_i r_i^n = 0 \quad (n = 0, 1, \dots, k-1),$$

a Vandermonde system with nonzero determinant, forcing $a_i = 0$ for all i . Thus $\{v_r : r \in \mathbb{R}\}$ is linearly independent, so $\dim(\mathbb{R}^{\mathbb{N}}) \geq |\mathbb{R}|$. Since $|\mathbb{R}^{\mathbb{N}}| = |\mathbb{R}|$, we get $\dim(\mathbb{R}^{\mathbb{N}}) = |\mathbb{R}|$ (uncountable). Hence $P(\mathbb{R}) \not\cong \mathbb{R}^{\mathbb{N}}$.

Theory — Bases, Subspaces, and Isomorphisms (for Problems 1–3)

A. Bases and change-of-basis matrices

Let $(e_1, \dots, e_n)$ be a basis of a real vector space V . Given vectors $(v_1, \dots, v_n)$ in V , write each v_j uniquely as

$$v_j = \sum_{i=1}^n a_{ij} e_i.$$

The coefficient matrix $A = (a_{ij})$ is the change-of-coordinates matrix from the e -basis to the v -list.

Basis criterion. The following are equivalent:

$$(v_1, \dots, v_n) \text{ is a basis} \iff \text{columns of } A \text{ are linearly independent} \iff A \text{ is invertible} \iff$$

Triangular case. If A is triangular (upper or lower), then

$$\det(A) = \prod_{i=1}^n a_{ii}.$$

Hence a triangular A with all diagonal entries nonzero is automatically invertible. In practice this corresponds to solving for the original basis vectors by back-substitution.

B. Subspaces: unions, sums, intersections, and inclusions

Subspace test. A nonempty subset $S \subseteq V$ is a subspace iff it is closed under addition and scalar multiplication.

Union of subspaces. If T and U are subspaces, then $T \cup U$ is a subspace only in the nested case:

$$T \cup U \text{ is a subspace} \iff T \subseteq U \text{ or } U \subseteq T.$$

Indeed, if $t \in T \setminus U$ and $u \in U \setminus T$, then $t + u$ cannot lie in T or U (otherwise subtraction would force containment), contradicting closure under addition.

Sum and intersection. For subspaces $T, U \leq V$,

$$T + U := \{t + u : t \in T, u \in U\}$$

is the smallest subspace containing T and U , and $T \cap U$ is always a subspace.

Always-true inclusions. For any subspaces $T, U, W \leq V$,

$$T + (U \cap W) \subseteq (T + U) \cap (T + W).$$

(If $x = t + z$ with $z \in U \cap W$, then $x \in T + U$ and $x \in T + W$.) Also,

$$(T \cap W) + (U \cap W) \subseteq (T + U) \cap W.$$

(Each summand lies in W and in $T + U$.)

Modular law (useful sufficient condition for equality). If $T \subseteq W$, then

$$T + (U \cap W) = (T + U) \cap W.$$

Thus “distributive-like” equalities may fail without extra containment hypotheses, but inclusions remain valid.

Grassmann formula (finite-dimensional V). If V is finite-dimensional and $T, U \leq V$, then

$$\dim(T + U) = \dim T + \dim U - \dim(T \cap U).$$

C. Vector-space isomorphisms

Definition. A linear map $L : V \rightarrow W$ is an isomorphism if it is bijective (equivalently, has a linear inverse).

Finite-dimensional classification. If V and W are finite-dimensional over $\mathbb{R}$, then

$$V \cong W \iff \dim V = \dim W.$$

Given bases (v_i) of V and (w_i) of W , the map $L(v_i) = w_i$ extends uniquely to an isomorphism.

Standard dimension facts. A single nontrivial linear equation in $\mathbb{R}^n$ defines a hyperplane of dimension $n - 1$. Polynomials of degree $\leq m$ have basis $\{1, x, \dots, x^m\}$ and hence dimension $m + 1$.

Function spaces via reparametrization. If $\alpha : [c, d] \rightarrow [a, b]$ is a homeomorphism, then

$$C[a, b] \cong C[c, d], \quad f \mapsto f \circ \alpha.$$

In suitable settings, integration/differentiation can yield isomorphisms, e.g.

$$C[0, 1] \rightarrow \{g \in C^1[0, 1] : g(0) = 0\}, \quad f \mapsto \left(x \mapsto \int_0^x f(t) dt\right).$$

Linear ODE solution spaces. For a linear homogeneous ODE of order k , the space of real solutions is a k -dimensional real vector space. For $\ddot{x} + x = 0$, a basis is $\{\cos t, \sin t\}$.

Infinite-dimensional obstructions. A finite-dimensional vector space is never isomorphic to an infinite-dimensional one. Moreover, $P(\mathbb{R})$ has a countable basis $\{1, x, x^2, \dots\}$, while $\mathbb{R}^{\mathbb{N}}$ has uncountable Hamel dimension (one can exhibit a continuum-sized linearly independent set via a Vandermonde argument). Therefore $P(\mathbb{R}) \not\cong \mathbb{R}^{\mathbb{N}}$.

Problem 4 — Sum of linear maps; images/kernels; idempotents

Exact statement.

- (i) If α and β are linear maps from U to V show that $\alpha + \beta$ is linear. Give explicit counter-examples to:

$$(a) \operatorname{im}(\alpha + \beta) = \operatorname{im}(\alpha) + \operatorname{im}(\beta), \quad (b) \operatorname{ker}(\alpha + \beta) = \operatorname{ker}(\alpha) \cap \operatorname{ker}(\beta).$$

Show that in general each equality can be replaced by a valid inclusion of one side in the other.

- (ii) Let $\alpha : V \rightarrow V$ be linear. Show that if $\alpha^2 = \alpha$ then $V = \operatorname{ker}(\alpha) \oplus \operatorname{im}(\alpha)$. Does your proof still work if V is infinite dimensional? Is the result still true?

Solution. (i) For $u, u' \in U$ and $\lambda \in \mathbb{R}$,

$$(\alpha + \beta)(u + u') = \alpha(u + u') + \beta(u + u') = (\alpha u + \alpha u') + (\beta u + \beta u') = (\alpha + \beta)u + (\alpha + \beta)u',$$

$$(\alpha + \beta)(\lambda u) = \alpha(\lambda u) + \beta(\lambda u) = \lambda \alpha(u) + \lambda \beta(u) = \lambda(\alpha + \beta)(u),$$

so $\alpha + \beta$ is linear.

Counterexample to (a): take $U = V = \mathbb{R}$, $\alpha = \operatorname{id}$, $\beta = -\operatorname{id}$. Then $\alpha + \beta = 0$, hence $\operatorname{im}(\alpha + \beta) = \{0\}$ but $\operatorname{im}(\alpha) + \operatorname{im}(\beta) = \mathbb{R}$. In general,

$$\operatorname{im}(\alpha + \beta) \subseteq \operatorname{im}(\alpha) + \operatorname{im}(\beta),$$

since $(\alpha + \beta)(u) = \alpha(u) + \beta(u)$.

Counterexample to (b): with the same α, β , we have $\operatorname{ker}(\alpha + \beta) = \mathbb{R}$ but $\operatorname{ker}(\alpha) \cap \operatorname{ker}(\beta) = \{0\}$. In general,

$$\operatorname{ker}(\alpha) \cap \operatorname{ker}(\beta) \subseteq \operatorname{ker}(\alpha + \beta),$$

since $\alpha(u) = \beta(u) = 0 \Rightarrow (\alpha + \beta)(u) = 0$.

- (ii) Assume $\alpha^2 = \alpha$. For any $x \in V$,

$$x = (x - \alpha x) + \alpha x.$$

Moreover,

$$\alpha(x - \alpha x) = \alpha x - \alpha^2 x = \alpha x - \alpha x = 0,$$

so $x - \alpha x \in \ker(\alpha)$ and $\alpha x \in \operatorname{im}(\alpha)$, hence $V = \ker(\alpha) + \operatorname{im}(\alpha)$.

If $y \in \ker(\alpha) \cap \operatorname{im}(\alpha)$, then $y = \alpha(x)$ for some x and $0 = \alpha(y) = \alpha(\alpha x) = \alpha^2 x = \alpha x = y$, so $y = 0$. Thus the sum is direct:

$$V = \ker(\alpha) \oplus \operatorname{im}(\alpha).$$

No finite-dimensional hypothesis was used, so the proof works for infinite-dimensional V and the result remains true.

Problem 5 — Bases for U , W , $U \cap W$, and a description of $U + W$ in $\mathbb{R}^5$

Exact statement. Let

$$U = \{x \in \mathbb{R}^5 : x_1 + x_3 + x_4 = 0, 2x_1 + 2x_2 + x_5 = 0\}, \quad W = \{x \in \mathbb{R}^5 : x_1 + x_5 = 0, x_2 = x_3 = x_4\}.$$

Find bases for U and W containing a basis for $U \cap W$ as a subset. Give a basis for $U + W$ and show that

$$U + W = \{x \in \mathbb{R}^5 : x_1 + 2x_2 + x_5 = x_3 + x_4\}.$$

Solution. For U , solve

$$x_1 = -x_3 - x_4, \quad x_5 = -2x_1 - 2x_2 = 2x_3 + 2x_4 - 2x_2.$$

With free variables x_2, x_3, x_4 ,

$$x = x_2(0, 1, 0, 0, -2) + x_3(-1, 0, 1, 0, 2) + x_4(-1, 0, 0, 1, 2).$$

Thus a basis of U is

$$u_1 = (0, 1, 0, 0, -2), \quad u_2 = (-1, 0, 1, 0, 2), \quad u_3 = (-1, 0, 0, 1, 2).$$

For W , write $x_2 = x_3 = x_4 = t$ and $x_1 = s$, so $x_5 = -s$. Then

$$x = (s, t, t, t, -s) = s(1, 0, 0, 0, -1) + t(0, 1, 1, 1, 0),$$

hence a basis of W is

$$w_1 = (1, 0, 0, 0, -1), \quad w_2 = (0, 1, 1, 1, 0).$$

For $U \cap W$, take $x = (s, t, t, t, -s) \in W$ and impose $x_1 + x_3 + x_4 = 0$:

$$s + 2t = 0 \Rightarrow s = -2t.$$

Therefore

$$U \cap W = \operatorname{span}\{v\}, \quad v = (-2, 1, 1, 1, 2).$$

Note that $v = u_1 + u_2 + u_3$ and $v = -2w_1 + w_2$, so bases containing v are

$$\mathcal{B}_U = \{v, u_2, u_3\} \text{ for } U, \quad \mathcal{B}_W = \{v, w_1\} \text{ for } W.$$

Since $\dim(U + W) = 3 + 2 - 1 = 4$, a basis for $U + W$ is

$$\mathcal{B}_{U+W} = \{v, u_2, u_3, w_1\}.$$

Let $S := \{x \in \mathbb{R}^5 : x_1 + 2x_2 + x_5 = x_3 + x_4\}$. If $x \in U + W$ and $x = av + bu_2 + cu_3 + dw_1$, then a coordinate check gives $x_1 + 2x_2 + x_5 = x_3 + x_4$, hence $U + W \subseteq S$.

Conversely, given $x \in S$, set

$$a = x_2, \quad b = x_3 - x_2, \quad c = x_4 - x_2, \quad d = x_1 + x_3 + x_4.$$

Then one verifies $x = av + bu_2 + cu_3 + dw_1$, hence $x \in U + W$, so $S \subseteq U + W$. Therefore $U + W = S$.

Problem 6 — Descending images, ascending kernels, and rank constraints

Exact statement.

- (i) Let $\alpha : V \rightarrow V$ be an endomorphism of a finite dimensional vector space V . Show that

$$V \supseteq \operatorname{im}(\alpha) \supseteq \operatorname{im}(\alpha^2) \supseteq \cdots, \quad \{0\} \subseteq \ker(\alpha) \subseteq \ker(\alpha^2) \subseteq \cdots.$$

If $r_k = \operatorname{rk}(\alpha^k)$, deduce that $r_k \geq r_{k+1}$ and $r_k - r_{k+1} \geq r_{k+1} - r_{k+2}$. Conclude that if for some $k \geq 0$ we have $r_k = r_{k+1}$, then $r_k = r_{k+\ell}$ for all $\ell \geq 0$.

- (ii) Suppose $\dim(V) = 5$, $\alpha^3 = 0$, but $\alpha^2 \neq 0$. What possibilities are there for $\operatorname{rk}(\alpha)$ and $\operatorname{rk}(\alpha^2)$?

Solution. (i) Since

$$\operatorname{im}(\alpha^{k+1}) = \alpha(\operatorname{im}(\alpha^k)) \subseteq \operatorname{im}(\alpha^k),$$

the images form a descending chain. Also $\alpha^k(v) = 0 \Rightarrow \alpha^{k+1}(v) = \alpha(0) = 0$, hence $\ker(\alpha^k) \subseteq \ker(\alpha^{k+1})$.

Let $r_k = \dim \operatorname{im}(\alpha^k)$. The inclusion $\operatorname{im}(\alpha^{k+1}) \subseteq \operatorname{im}(\alpha^k)$ implies $r_k \geq r_{k+1}$.

Consider the restriction $\alpha : \operatorname{im}(\alpha^k) \rightarrow \operatorname{im}(\alpha^{k+1})$, which is surjective. Rank-nullity for this restricted map gives

$$\dim(\ker(\alpha) \cap \operatorname{im}(\alpha^k)) = \dim \operatorname{im}(\alpha^k) - \dim \operatorname{im}(\alpha^{k+1}) = r_k - r_{k+1}.$$

Similarly,

$$r_{k+1} - r_{k+2} = \dim(\ker(\alpha) \cap \operatorname{im}(\alpha^{k+1})).$$

Since $\operatorname{im}(\alpha^{k+1}) \subseteq \operatorname{im}(\alpha^k)$, we have

$$\ker(\alpha) \cap \operatorname{im}(\alpha^{k+1}) \subseteq \ker(\alpha) \cap \operatorname{im}(\alpha^k),$$

so $r_{k+1} - r_{k+2} \leq r_k - r_{k+1}$, i.e.

$$r_k - r_{k+1} \geq r_{k+1} - r_{k+2}.$$

If $r_k = r_{k+1}$, then $r_k - r_{k+1} = 0$, hence $0 \geq r_{k+1} - r_{k+2}$. But also $r_{k+1} \geq r_{k+2}$, so $r_{k+1} = r_{k+2}$. Inductively, $r_k = r_{k+\ell}$ for all $\ell \geq 0$.

(ii) Here $r_3 = 0$ and $r_2 \geq 1$. Also $\text{im}(\alpha^2) \subseteq \ker(\alpha)$ because $\alpha(\alpha^2(v)) = \alpha^3(v) = 0$. Thus

$$r_2 \leq \dim \ker(\alpha) = 5 - r_1 \Rightarrow r_1 + r_2 \leq 5.$$

From (i), the successive drops satisfy

$$(5 - r_1) \geq (r_1 - r_2) \geq (r_2 - 0) = r_2,$$

so $r_1 \geq 2r_2$ and $r_2 \geq 1$. The only integer solutions are

$$(\text{rk}(\alpha), \text{rk}(\alpha^2)) = (2, 1) \quad \text{or} \quad (3, 1).$$

Both occur (Jordan types $3 + 1 + 1$ and $3 + 2$ respectively).

Theory — Linear maps, projections, and powers (Problems 4–6)

4. Sums of linear maps; images and kernels

Let U, V be vector spaces over $\mathbb{R}$. The set

$$\text{Hom}(U, V) = \{\text{linear maps } U \rightarrow V\}$$

is a vector space with operations defined pointwise:

$$(\alpha + \beta)(u) = \alpha(u) + \beta(u), \quad (c\alpha)(u) = c\alpha(u).$$

Hence if α, β are linear then $\alpha + \beta$ is linear.

Always-true inclusions. For linear maps $\alpha, \beta : U \rightarrow V$,

$$\text{im}(\alpha + \beta) \subseteq \text{im}(\alpha) + \text{im}(\beta),$$

since $(\alpha + \beta)(u) = \alpha(u) + \beta(u)$ is a sum of an element of $\text{im}(\alpha)$ and an element of $\text{im}(\beta)$. Also

$$\ker(\alpha) \cap \ker(\beta) \subseteq \ker(\alpha + \beta),$$

since $\alpha(u) = \beta(u) = 0 \Rightarrow (\alpha + \beta)(u) = 0$. Equalities may fail because of cancellation (e.g. $\beta = -\alpha$).

4(ii). Idempotents and projections

A linear map $\alpha : V \rightarrow V$ is *idempotent* if $\alpha^2 = \alpha$. Then α is a projection onto its image along its kernel:

Decomposition. For any $x \in V$,

$$x = (x - \alpha x) + \alpha x,$$

and

$$\alpha(x - \alpha x) = \alpha x - \alpha^2 x = \alpha x - \alpha x = 0,$$

so $x - \alpha x \in \ker(\alpha)$, while $\alpha x \in \text{im}(\alpha)$. Thus $V = \ker(\alpha) + \text{im}(\alpha)$.

Directness. If $y \in \ker(\alpha) \cap \text{im}(\alpha)$, then $y = \alpha(x)$ for some x and

$$0 = \alpha(y) = \alpha(\alpha x) = \alpha^2 x = \alpha x = y,$$

so $y = 0$ and

$$V = \ker(\alpha) \oplus \text{im}(\alpha).$$

No dimension hypothesis is used, so the proof works and the statement remains true even if V is infinite-dimensional.

5. Subspaces defined by linear equations

A subset of $\mathbb{R}^n$ given by homogeneous linear equations,

$$U = \{x \in \mathbb{R}^n : Ax = 0\},$$

is a subspace (it is nonempty and closed under addition and scalar multiplication).

Bases via parametrization. To find a basis of such a subspace, solve the system, choose free parameters, and express the general solution as a linear combination of parameter-vectors. These parameter-vectors span the solution space and are linearly independent when they correspond to independent parameters.

Intersections and sums. Intersections $U \cap W$ are found by solving both systems simultaneously. Sums $U + W$ are subspaces, and in finite dimensions one has the Grassmann formula:

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

A practical way to build a basis for $U + W$ is: start with a basis of $U \cap W$, extend to bases of U and W , then combine and discard redundant vectors.

6. Powers of an endomorphism; rank stabilization; nilpotent patterns

Let $\alpha : V \rightarrow V$ be linear. Then for all $k \geq 0$,

$$\operatorname{im}(\alpha^{k+1}) = \alpha(\operatorname{im}(\alpha^k)) \subseteq \operatorname{im}(\alpha^k), \quad \ker(\alpha^k) \subseteq \ker(\alpha^{k+1}),$$

so images form a descending chain and kernels form an ascending chain.

Ranks and drops. Let $r_k = \operatorname{rk}(\alpha^k) = \dim \operatorname{im}(\alpha^k)$. Then $r_k \geq r_{k+1}$. Moreover, the restriction

$$\alpha : \operatorname{im}(\alpha^k) \rightarrow \operatorname{im}(\alpha^{k+1})$$

is surjective, so rank-nullity yields

$$r_k - r_{k+1} = \dim(\ker(\alpha) \cap \operatorname{im}(\alpha^k)).$$

Since $\operatorname{im}(\alpha^{k+1}) \subseteq \operatorname{im}(\alpha^k)$, we have

$$\ker(\alpha) \cap \operatorname{im}(\alpha^{k+1}) \subseteq \ker(\alpha) \cap \operatorname{im}(\alpha^k),$$

hence

$$r_k - r_{k+1} \geq r_{k+1} - r_{k+2}.$$

In particular, if $r_k = r_{k+1}$ for some k , then all subsequent ranks stabilize: $r_k = r_{k+\ell}$ for every $\ell \geq 0$.

Nilpotent operators. If α is nilpotent, it has a Jordan decomposition with block sizes $\lambda_1, \dots, \lambda_t$. Then

$$\operatorname{rk}(\alpha) = \sum_{i=1}^t (\lambda_i - 1), \quad \operatorname{rk}(\alpha^2) = \sum_{i=1}^t \max(\lambda_i - 2, 0).$$

Thus rank constraints for α, α^2 can be read off from possible partitions of $\dim V$ into allowed block sizes.